TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Durable Anion Exchange Polymer Membrane and Ionomer Binder for Iridium-Free AEM Water Electrolyzers			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	New/renewable energy	Hydrogen-manufacturing	High polymer materials	Material technology for energy and environmental industry purpose
Concept of Proposed Technology	**Concept and Background** Among many alternative energy technologies, hydrogen based electrochemical devices have gained significance recently because they do not involve environmentally harmful byproducts or reactants. Water electrolyzers split water into hydrogen and oxygen using electricity (preferably from sustainable source, such as solar and wind energy). Once H_2 from water electrolyzer is stored, it can be used to generate electrical energy via fuel cells while releasing only water as a byproduct. State-of-the-art hydrogen fuel cells and electrolyzers employ proton exchange membranes (PEMs) as a solid electrolyte operating in acidic media. Unfortunately, these technologies suffer from several drawbacks including the requirement of platinum metal group electrocatalysts and durable corrosion resistant bipolar plates at low pH. [1]			

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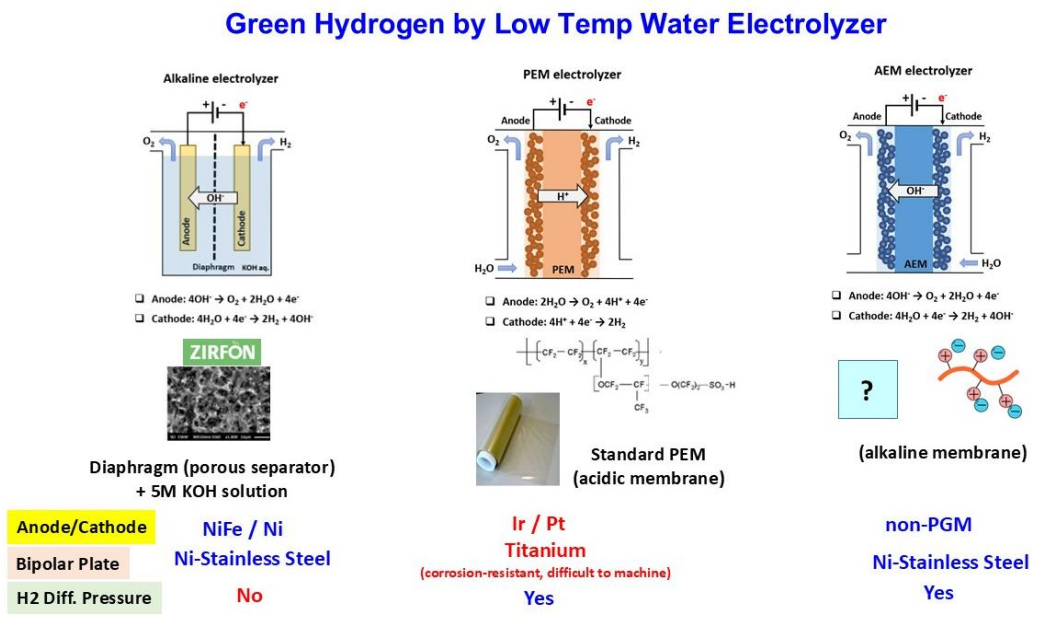

Figure 1. Different types of low temperature (<100 °C) water electrolyzer.

Core technical gap (problem statement)

Anion exchange membrane (AEM) water electrolyzer represents a promising technology that combines advantages of both liquid alkaline electrolyzer and PEM water electrolyzer (see Figure 1).[2] Similar to liquid alkaline electrolyzers, AEM water electrolyzer operates under alkaline environments but uses an alkaline membrane (i.e., AEM) to transport OH^- ion across the electrodes instead of a porous diaphragm separator. AEM water electrolyzer can also use nonprecious metals, such as nickel or iron, as electrocatalysts and stainless-steel bipolar plates while PEM water electrolyzer requires precious metal catalysts and titanium bipolar plates. Despite these advantages, full potential of AEM electrolyzer has not been materialized, in part due to the lack of low-cost, easily scalable, durable, high-performance AEM materials that can serve as a standard AEM water electrolyzer.

RPI Innovation

Bae group of RPI has developed a highly ion conductive, chemically and mechanically stable, easily manufacturable AEM that can be used key materials for production of green hydrogen by AEM water electrolyzer.[3] Among over 50 AEM samples independently tested by the National Renewable Laboratory (NREL) in US, two previous generations of RPI membranes have already demonstrated the best performance in ion conductivity and durability in 1M KOH solution at 80 C over 1,000 h test (PPN4 and PPN5 of Figure 2).[4] One of them was commercialized as Orion TM1™ by Orion Polymer Corp. and has been recognized as one of the most durable high-performance AEMs by many electrolyzer experts globally so far.[5]

In this TIPA collaboration project with a SME, the RPI team will extend our AEM work to further improve conductivity, lower cost, and enhance mechanical strength by creating well-defined and interconnected ion channels through “channel engineering” of the hydrophobic/hydrophilic

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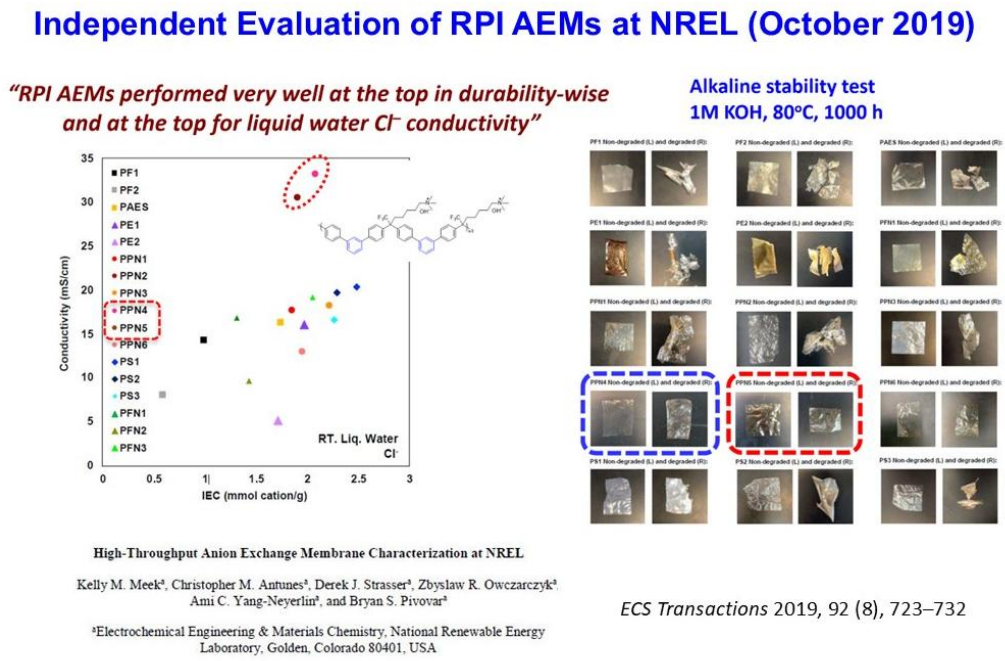
	morphology. The highly tunable polymer material platform of RPI invention allows development of new ion conducting polymer materials that can be used as an AEM (i.e., membrane) and anion exchange ionomer (i.e., catalyst binder) with appropriate chemical manipulations of polymeric backbone and sidechain ionic group. Those novel anion exchange membrane and ionomer materials will be evaluated in AEM water electrolyzer MEAs and stacks by a SME who specializes in electrode engineering and stack engineering.  Figure 2. Independent performance and stability verification of RPI AEMs by NREL.
Rationale for Joint R&D Project	Although AEM water electrolyzer offers enormous potential for low-cost green hydrogen production, its water electrolyzer system is composed of various parts that need further engineering of each component and seamless integration into electrolyzer membrane electrode assembly (MEA) and stack. It is difficult for one organization alone to cover entire technology areas. The RPI team has global reputation for high-performance AEM polymer materials development and is looking for a SME collaborator who can provide complementary expertise in MEA and stack engineering and lead the global AEM water electrolyzer market. Despite that the current RPI AEM materials have shown good performance and durability in ex-situ conditions, their integration into catalyst coated membrane and stack level requires further optimization in design and properties of AEM materials based on the feedback from MEA and stack tests. Thus, multiple iterations of material development, cell/stack test, and feedback are critical for successful development of AEM electrolyzer system. Synergistic effect of this collaboration would result in development of durable, low-cost, potentially platinum group metal (PGM)-free AEM water electrolyzer that could lead global market of green hydrogen production technology (Figure 3).

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	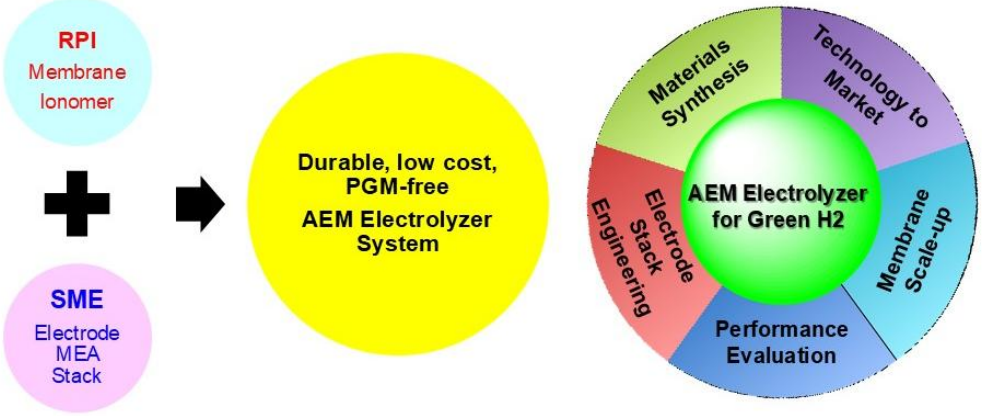 Figure 3. Integrated collaboration approach for AEM water electrolyzer development.
Key Words	1. green hydrogen 2. water electrolyzer 3. anion exchange membrane 4. MEAs and stack engineering

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 5

R&D Objectives

1. RPI: Develop and optimize hydroxide conducting anion exchange polymers for membrane and ionomer catalyst binder uses of AEM water electrolyzer MEAs and stack.
2. SME: Evaluate the RPI polymer materials in MEAs and stack level and suggest outcomes of materials' properties in cell test
3. RPI and SME: Through multiple iterations of RPI materials test-and-feedback between RPI and SME, optimize materials, electrodes, and other hardware components of AEM water electrolyzer stack and establish a structure–property map and guidance on scale-up.
4. SME: Demonstration of (a) AEM water electrolyzer performance and durability at stack level and (b) its commercial potential in green hydrogen market.

R&D Target Specifications

	Performance Indicator	Unit	Target	Global Best	Standard/ Certification Criteria
1	OH^- conductivity	mS/cm	120	100 (RPI)	In-plane hydroxide conductivity in liquid water at 80 °C
2	Tensile strength	MPa	≥35	30 (RPI)	Tensile strength measurement at 50 °C and 50% relative humidity
3	Swelling ratio	%	≤12%	20% (RPI)	Swelling ratio measurement in x- and y-direction after soaking into water for 12 h
4	Stack Performance	V	1.8	1.8 (Enapter)	Cell-average voltage at j = 0.6 $A \cdot cm^{-2}$ in 0.3 M KOH at 60 °C
5	Stack Durability	%	10	10 (Enapter)	Voltage increase ratio ≤ 10% after 1,000 h at j = 0.6 $A \cdot cm^{-2}$ in 0.3 M KOH at 60 °C (baseline = initial stabilized voltage at same condition)

R&D Plan

Task	Period (Month)	Key Tasks & Methodology	Main Deliverables / Milestones
Task 0	0–36	Project governance (joint meeting, reporting), IP/contracting, EHS, quality management (QC)	Kick-off meting; protocol & data management plan; periodic reports; IP strategy; final technical report.
Task 1 SME	0–6	Basic property characterization of current generation RPI polymers materials (membrane and ionomer binder) in MEA test at SME for benchmarking. Optimization of MEA test conditions (temperature, KOH concentration, etc).	Benchmarking report; formulation of Phase 2 plan.
Task 2 RPI	3–18	Anion exchange polymer chemistry design: select appropriate combination of backbone and sidechain ionic group for controls of molecular weight, flexibility by tuning hard/soft backbone; try different quaternary ammonium groups. Characterize relationships of polymer chemical structure and mechanical property, thermal property, and membrane property (IEC, swelling).	Down-selection of 1st-gen polymer candidates (≥2); structure–property map by linking chemistry variables to membrane properties and MEA test.
Task 3 SME	7–24	MEA & stack screening of RPI 1st-gen candidates (iterative feedback loop)	MEA performance ranking report (≥2 candidates);

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			1) Establish fixed test matrix for MEA (e.g., 0.3 M KOH, 60 °C, j = 0.6 A·cm⁻²) and evaluate the cell performance. 2) Translate best candidates to stack-level tests (single-cell → short stack) to verify scalability and identify failure modes; feed results back to RPI for chemistry iteration.	validated MEA/stack test protocol; selection of candidate(s) for Task 4 optimization.
	Task 4 RPI	19–30	Optimizations of polymer synthetic process and membrane casting of down-selected 1st-gen candidates: molecular weight control, solvents selection, isolation process, membrane casting parameter, thickness optimization.	Down-select 1 polymer material candidate. Delivery of thickness-optimized down-selected AEM ≥5 pieces in 10cm x 10cm size and ionomer binder solution ≥100 mL for MEA and stack tests.
	Task 5 SME	25–36	Final validation of down-selected polymer in stack/system conditions. 1) Confirm performance target at j = 0.6 A·cm⁻² in 0.3 M KOH, 60 °C (cell-average voltage). 2) Execute 1,000 h durability test at the same condition and quantify voltage increase ratio. 3) Demonstrate stack integration readiness (assembly compatibility, sealing, operability)	Stack performance report; 1,000 h durability report (≤10% voltage increase);
	Task 6 RPI & SME	30–36	Scale up and commercialization planning: cost model, supply chain, production scale-up roadmap, and customer engagement for adoption.	Go-to-market plan; pilot production plan; partner/customer LOI (if available).

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Polymer material (AEM and ionomer) design, synthesis, and characterization: Optimize polymer architecture and synthetic chemistry to balance ion conductivity, mechanical property, and processability.
	• Advanced characterization of polymers by NMR, FT-IR, viscoelasticity (DMA/rheology), morphology, thermal property (DSC/SEM as needed). • Failure analysis of membrane and ionomer: identify potential degradation mechanisms after integration into MEAs and durability test.
Expected R&R of Korean SME Partner	• Electrode and stack engineering based on RPI AEM and ionomer. • Establish and execute MEA/stack test protocols and benchmarking: define operating conditions (e.g., 0.3 M KOH, 60 °C, j = 0.6 A·cm^{-2}) and evaluate performance and stability. • Conduct durability validation and failure feedback loop • Run long-term tests (e.g., 1,000 h) at the defined condition, quantify voltage increase, and provide test data/failed samples to RPI for degradation analysis and iteration.
	• Secure intellectual property. • Lead commercialization: product spec, cost/production plan, customer expansion, and regulatory/safety documentation.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Outcomes:** • Engineering optimization of multiple materials components and manufacturing processes of AEM water electrolyzer system: membrane, ionomer, catalysts, electrodes, catalyst coating process, MEA fabrication, gasket design, stack assembly. • Mapping of failure-mode of AEM water electrolyzer MEA and stack and establishment of guideline of prevention protocols. • Demonstration of durable, high-performance, iridium-free AEM water electrolyzer short-stack operation using 30 kW-class large-area electrodes (> 500 cm^2) for the 1st time in the world. **Impact:** • Accelerate the commercial potential of AEM water electrolyzers, driving down the cost of green hydrogen production. • Establish a secure global supply chain for key materials of membrane ionomer that have potential of use in global market of green hydrogen production through joint IP development. • Boost the global visibility and market share of Korean SMEs in green hydrogen sector by creating robust international value chains.

6. References

[1] Kim, Y. S. (2025). "Making hydrogen production durable." Science **390**(6770): 237-238.
[2] Park, E. J., et al. (2024). "Aryl ether-free polymer electrolytes for electrochemical and energy devices." Chemical Society Reviews **53**(11): 5704-5780.
[3] Lee, W.-H., et al. (2015). "Robust Hydroxide Ion Conducting Poly(biphenyl alkylene)s for Alkaline Fuel Cell Membranes." ACS Macro Letters: 814-818.
[4] Meek, K. M., et al. (2019). "High-Throughput Anion Exchange Membrane Characterization at NREL." ECS Transactions **92**(8): 723-731.
[5] Noh, S., et al. (2019). "Molecular Engineering of Hydroxide Conducting Polymers for Anion Exchange Membranes in Electrochemical Energy Conversion Technology." Accounts of Chemical Research **52**(9): 2745-2755.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization